

The Work of Reducing Dependence

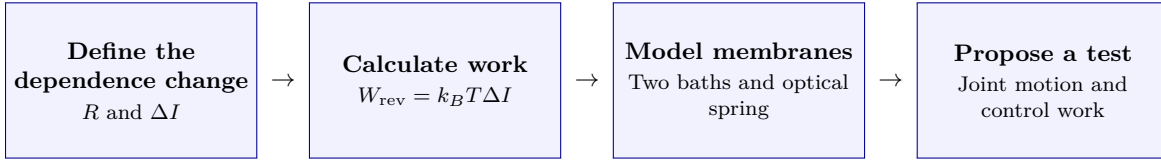
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Abstract

Can we predict the work needed to make two physical systems less dependent, then check the prediction against an experiment? We begin with the pointwise dependence ratio $R = p_{12}/(p_1 p_2)$ and use its average log, mutual information I , to measure overall dependence. A solvable equilibrium model predicts exactly that a slow reduction of I by ΔI costs $k_B T \Delta I$ of work. We then examine two cavity-coupled membranes as a possible test of the same dependence-and-work question. Their published coupling and heat-flux measurements constrain the model, but the real device has two baths and an optical spring that changes each membrane's frequency, so its work cannot be inferred from ΔI alone. In that model, lowering coupling reduces position dependence while increasing dependence carried by the heat current. It still requires positive control work. Existing figures check the heat-flux scale but do not report the joint motion and control work needed to test these predictions. We state the required measurements explicitly.

From a mathematical dependence change to an experimental test



1 Dependence and work

Let x and y denote simultaneous position measurements of two oscillators. Their pointwise dependence ratio is

$$R(x, y) = \frac{p(x, y)}{p_1(x)p_2(y)}, \quad I = \langle \log R \rangle = \iint p(x, y) \log R(x, y) dx dy. \quad (1)$$

If the positions are independent, $R = 1$ almost everywhere and $I = 0$. For dependent positions, R may be above one in some regions and below one in others. The average in Eq. (1) is their Kullback–Leibler divergence from the product distribution, in nats. We use it to compare overall dependence while retaining R to locate changes within the state space.

For a chosen pair of observables, define the before-to-after change as $\Delta I = I_{\text{before}} - I_{\text{after}}$. A positive ΔI means that the protocol reduces their dependence. The question is how much work W_{on} must be supplied to produce that change. The ratio R describes each state, but work depends on the controlled Hamiltonian, baths, and path between states. We first answer the question exactly in a one-bath model, then apply the same definitions to a membrane system that needs a different work calculation.

2 Equilibrium prediction

To calculate a work cost for the ΔI defined above, we start with two classical oscillators in one bath at temperature T . Their controlled potential is

$$V_{\times}(x, y) = \frac{k}{2}(x^2 + y^2) + gxy, \quad q = g/k, \quad |q| < 1. \quad (2)$$

The Gibbs distribution has covariance $k_B T \begin{pmatrix} k & g \\ g & k \end{pmatrix}^{-1}$. Its position correlation coefficient is $\rho = -q$. Gaussian integration then gives

$$\rho = -q, \quad I(q) = -\frac{1}{2} \log(1 - q^2), \quad W_{\text{rev}} = k_B T [I(q_{\text{before}}) - I(q_{\text{after}})]. \quad (3)$$

We define positive W_{rev} as work done *on* the oscillators. The equality holds for quasistatic isothermal control because $F(q) - F(0) = (k_B T/2) \log(1 - q^2) = -k_B T I(q)$ (Appendix A). As $|q|$ decreases, I falls and F rises. The relation depends on this free-energy function; another Hamiltonian need not obey it.

Yang *et al.* report two membrane modes near $\omega_0/2\pi = 400$ kHz and tunable cavity-mediated coupling Λ .^[1] Their measured normal-mode splitting implies $|q| \simeq 2|\Lambda|/\omega_0$ for the cross-coupling term. The earlier estimate $|q| \simeq |\Lambda|/\omega_0$ missed a factor of two. Using the corrected value, a change from $|\Lambda|/2\pi = 85$ to 58 Hz in the *equal-temperature cross-coupling model* would give $\Delta I = 4.83 \times 10^{-8}$ nats and $W_{\text{rev}} = 2.00 \times 10^{-28}$ J at 300 K. These numbers establish a benchmark for the same before-and-after coupling values; they are not the work cost of the membrane apparatus.

3 Modeling the membranes

We now ask what happens to dependence and work for that same 85 to 58 Hz change in the membranes. Two features prevent us from using Eq. (3) as their work prediction. One membrane is driven by added white noise while the other contacts a room-temperature bath, producing a nonequilibrium steady state with heat flow.^[1] The cavity also shifts each mode's frequency along with their coupling. An effective classical Hamiltonian consistent with the reported mode matrix to first order in $|\Lambda|/\omega_0$ is

$$H_{\text{opt}} = \frac{p_1^2 + p_2^2}{2m} + \frac{m\omega_0^2}{2}(x_1^2 + x_2^2) + m\omega_0\Lambda(x_1 + x_2)^2. \quad (4)$$

Writing $h = 2\Lambda/\omega_0$, its stiffness matrix is $m\omega_0^2 \begin{pmatrix} 1+h & h \\ h & 1+h \end{pmatrix}$. The normal frequencies are ω_0 and $\omega_0\sqrt{1+2h}$, reproducing the observed splitting to first order. At equal temperature its free-energy change is

$$F_{\text{opt}}(h) - F_{\text{opt}}(0) = \frac{k_B T}{2} \log(1 + 2h). \quad (5)$$

For the slow 85 to 58 Hz change, Eq. (5) gives 5.60×10^{-25} J at 300 K, about 2800 times the cross-only benchmark. Even at equal temperature, the same change in coupling has a different work cost because the local spring shifts in Eq. (4) contribute at first order in h , while position dependence changes only at second order (Appendix A).

With two baths, we calculate the joint fluctuations generated by Eq. (4). The definition in Eq. (1) still applies when x and y stand for full oscillator states, but we must specify which observable pair is becoming more or less dependent. We use R_{pos} for the positions and R_{ph} for the full oscillator states $(x_1, \dot{x}_1)$ and $(x_2, \dot{x}_2)$. Their average logs are I_{pos} and I_{ph} . The latter includes the position-velocity correlation associated with heat transport. For resonant modes, weak optical-spring shifts, and independent Gaussian bath forces, a rotating-wave calculation links it to the mean heat current J :

$$I_{\text{ph}} = -\log(1 - r_J^2), \quad r_J = \frac{|J|}{2|\Lambda|k_B\sqrt{T_1^{\text{eff}}T_2^{\text{eff}}}}. \quad (6)$$

The dimensionless r_J is the magnitude of the normalized cross-correlation between mode amplitudes. Equation (6) can therefore estimate I_{ph} from J , Λ , and the two effective temperatures. A measured joint covariance would test both the pointwise ratio and its average more directly, without relying on the rotating-wave approximation (Appendix B).

For scale, take $T_H/T_L = 430$, approximately the hot-bath level in the published temperature plot. We solve the full linear Langevin covariance equation, including terms omitted from Eq. (6). At the two reported coupling values it gives

Predicted quantity	85 Hz	58 Hz
I_{pos} (nats)	9.05×10^{-8}	4.22×10^{-8}
I_{ph} (nats)	0.00123	0.00264
Mean heat current (W)	4.45×10^{-17}	4.44×10^{-17}

Using the before-minus-after convention of Section 1, $\Delta I_{\text{pos}} = 4.83 \times 10^{-8}$ nats, whereas $\Delta I_{\text{ph}} = -1.41 \times 10^{-3}$ nats. The positions become less dependent, but the full oscillator states become more dependent. A weaker coupling sustains a larger temperature difference, and the position–velocity correlation responsible for heat transport becomes more prominent relative to each oscillator’s fluctuations. Thus even the phrase *reduce the correlation* needs an observable attached to it.

Figure 1 shows the underlying pointwise ratios for two pairs of observables. Within each row the same color scale is used at both coupling settings. The position ratio moves closer to one as coupling falls; the position–velocity ratio moves further from one. The ratio does not rise or fall at every point. Mutual information summarizes its average log, not a pointwise ordering.

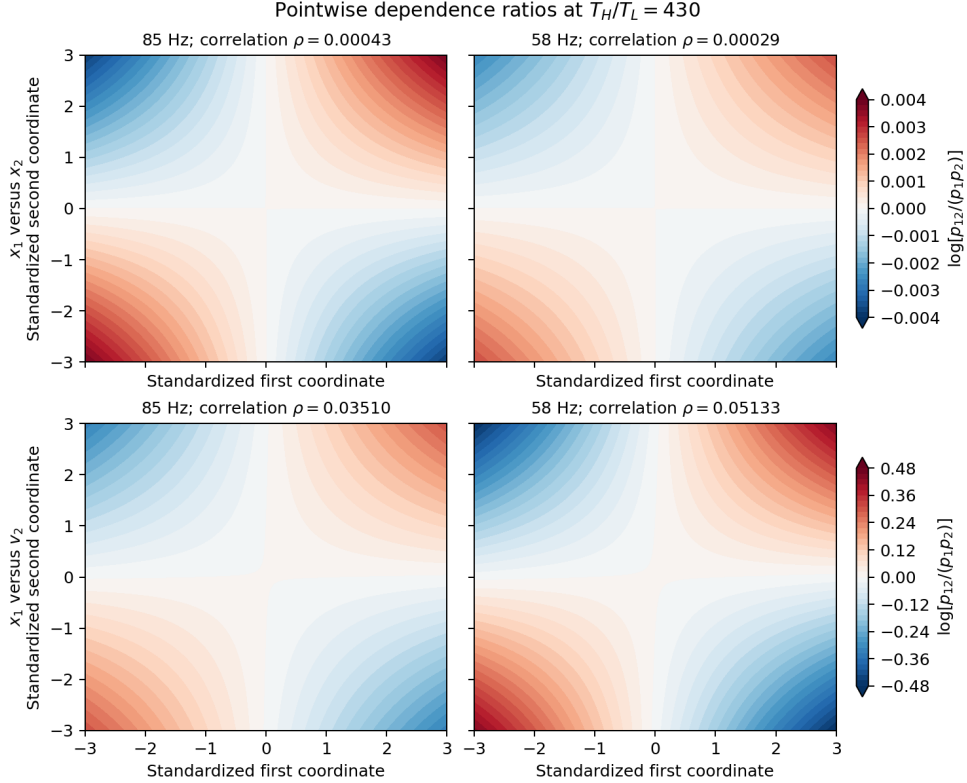


Figure 1: Predicted pointwise log ratios for standardized (x_1, x_2) and $(x_1, \dot{x}_2)$ readings. These are two-variable marginals of the model, not reported experimental histograms.

Having identified the dependence changes, we calculate work for the same coupling protocol. Equation (3) does not apply to this driven optical spring; its controlled Hamiltonian instead gives

$$W_{\text{on}} = m\omega_0 \int \dot{\Lambda}(t) \langle [x_1(t) + x_2(t)]^2 \rangle dt. \quad (7)$$

The average in Eq. (7) must be known *throughout* the protocol. We evolve the Gaussian covariance during linear 85 to 58 Hz ramps lasting 0.1 ms to 1 s, each initialized in the 85 Hz steady state (Appendix C). All ramps require positive work on the *effective mechanical Hamiltonian*. For $T_H/T_L = 430$, the 1 s result is 1.60×10^{-22} J at $T_L = 300$ K. It exceeds the cross-only benchmark because local spring terms contribute at first order in Λ and the hot bath increases fluctuations. It excludes laser power and heat spent maintaining the steady state; the experiment did not report this ramp. Figure 2 separates total work from the smaller effect of ramp duration.

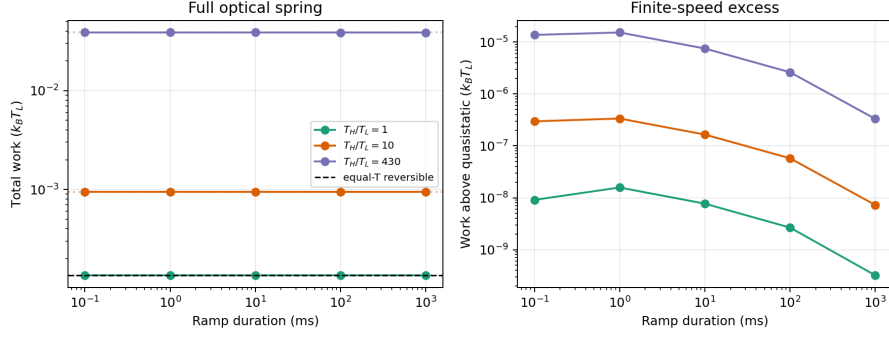


Figure 2: Predicted work for the 85 to 58 Hz change. Total work (left) is dominated by the optical spring’s local frequency shift. The right panel subtracts the integral of the steady-state generalized force along the same path, exposing finite-speed excess. This reference is a free-energy difference only when the baths have equal temperatures. None of these curves are measured apparatus power.

4 What published data can test

The published normal-mode splitting and heat-flux oscillation frequency support the coupling scale $2|\Lambda|$. [1] Its mean heat flux approaches roughly 4×10^{-17} W; using the approximate bath ratio above, the model predicts a saturation scale of 4.46×10^{-17} W. This is an order-of-magnitude check, not a fit to source data. The published temperature curves also approach one another as coupling grows, as the model predicts qualitatively. Their high-power values appear below our simple model’s $\simeq 287 T_L$ prediction, so that comparison remains unresolved.

The public article does not supply the synchronized joint motion data or a calibrated 85 to 58 Hz work protocol needed to test the predicted dependence and work changes. Its data and code are available from the authors on request. [1] The next comparison is specific: reconstruct R_{pos} and R_{ph} from both membrane quadratures at several couplings, calculate their average logs, and record the control trajectory and energy flows during a coupling change. The measured values can then be compared with Eqs. (6) and (7) using the same settings.

5 Conclusion

The theory gives an exact work–information prediction for a clean equilibrium benchmark. Asking the same before-and-after question for the membranes reveals a sharper test: identify *which* dependence is being reduced, include the local optical spring in the work calculation, and measure both the joint state and control energy. Published results check parts of the transport model, but they do not yet test the predicted dependence and work changes. Entanglement belongs to a separate quantum version of this question.

Reproducibility. Run `optical_spring_sweep.py` for the finite-time CSV and figures.

Run `extensions_calculations.py` for the equilibrium and steady-state values. The equations and covariance method are detailed in the earlier working notes and the experiment’s Supplementary Information. [2]

A Equilibrium calculations

For Eq. (2), the stiffness and position covariance matrices are

$$K_{\times} = k \begin{pmatrix} 1 & q \\ q & 1 \end{pmatrix}, \quad C_{\times} = k_B T K_{\times}^{-1} = \frac{k_B T}{k(1-q^2)} \begin{pmatrix} 1 & -q \\ -q & 1 \end{pmatrix}. \quad (8)$$

The kinetic partition function is independent of q , whereas the position partition function is proportional to $(\det K_{\times})^{-1/2}$. Hence $F_{\times}(q) - F_{\times}(0) = (k_B T/2) \log(1 - q^2) = -k_B T I(q)$, proving Eq. (3).

For any zero-mean Gaussian pair standardized to unit variance, with correlation coefficient ρ , the pointwise ratio is

$$\log R_\rho(a, b) = -\frac{1}{2} \log(1 - \rho^2) + \frac{\rho ab - \frac{1}{2} \rho^2 (a^2 + b^2)}{1 - \rho^2}. \quad (9)$$

This is the function shown in Fig. 1; averaging it over the joint Gaussian gives $I = -\frac{1}{2} \log(1 - \rho^2)$. Its sign can change across (a, b) even when $I > 0$.

For Eq. (4), let $h = 2\Lambda/\omega_0$. Then $K_{\text{opt}} = m\omega_0^2 \begin{pmatrix} 1+h & h \\ h & 1+h \end{pmatrix}$, with determinant $(m\omega_0^2)^2(1+2h)$. Its equilibrium position correlation is $\rho_{\text{pos}} = -h/(1+h)$, so

$$F_{\text{opt}}(h) - F_{\text{opt}}(0) = \frac{k_B T}{2} \log(1 + 2h), \quad I_{\text{pos}}(h) = -\frac{1}{2} \log \left[1 - \left(\frac{h}{1+h} \right)^2 \right]. \quad (10)$$

For small $|h|$, the work change is first order in the change of h , while $I_{\text{pos}} = h^2/2 + O(h^3)$. This explains why the optical-spring work cannot be inferred from Eq. (3).

B Steady-state covariance

The experiment's resonant effective-mode equation has a coupling Λ both on and off its diagonal.[1] In the rotating-wave approximation, define bath occupations $n_i = k_B T_i / (\hbar \omega_0)$ and $s = \text{Im} \langle b_1 b_2^* \rangle$. The steady solution is

$$s = \frac{2\Lambda\gamma_1\gamma_2(n_1 - n_2)}{(\gamma_1 + \gamma_2)(\gamma_1\gamma_2 + 4\Lambda^2)}, \quad N_{11} = n_1 - \frac{2\Lambda s}{\gamma_1}, \quad N_{22} = n_2 + \frac{2\Lambda s}{\gamma_2}, \quad J = 2\hbar\omega_0\Lambda s. \quad (11)$$

The real cross-correlation vanishes at exact resonance. Writing $r_J = |s|/\sqrt{N_{11}N_{22}}$ gives Eq. (6). The phase-space mutual information is $-\log(1 - r_J^2)$ because the two quadrature pairs have the same correlation magnitude.

For the numerical calculations we retain the full position and velocity dynamics. In dimensionless coordinates $z = (u_1, v_1, u_2, v_2)$, with $u_i = \omega_0 x_i / \sqrt{k_B T_L / m}$ and $v_i = \dot{x}_i / \sqrt{k_B T_L / m}$, their covariance satisfies

$$\dot{C} = A(h)C + CA(h)^\top + D, \quad A(h) = \begin{pmatrix} 0 & \omega_0 & 0 & 0 \\ -\omega_0(1+h) & -\gamma_1 & -\omega_0 h & 0 \\ 0 & 0 & 0 & \omega_0 \\ -\omega_0 h & 0 & -\omega_0(1+h) & -\gamma_2 \end{pmatrix}, \quad D = \text{diag}(0, 2\gamma_1 T_H / T_L, 0, 2\gamma_2). \quad (12)$$

The steady state solves $AC + CA^\top + D = 0$. Partition C into the two oscillator blocks C_1, C_2 . Gaussian integration then gives

$$I_{\text{ph}} = \frac{1}{2} \log \frac{\det C_1 \det C_2}{\det C}, \quad \log R_{\text{ph}}(z) = I_{\text{ph}} - \frac{1}{2} z^\top [C^{-1} - \text{diag}(C_1^{-1}, C_2^{-1})] z. \quad (13)$$

The same formulas applied to the 2×2 position submatrix give I_{pos} and R_{pos} . These definitions show why two distributions with different observed variables can have opposite dependence trends.

C Finite-time covariance and work

We divide each linear ramp into 400 intervals with constant midpoint h . If $C_{\text{ss}}(h)$ solves the steady Lyapunov equation, one interval of duration δt updates the covariance exactly:

$$C(t + \delta t) = C_{\text{ss}}(h) + e^{A(h)\delta t} [C(t) - C_{\text{ss}}(h)] e^{A(h)^\top \delta t}. \quad (14)$$

The work increment follows Eq. (7), or equivalently

$$\frac{dW_{\text{on}}}{k_B T_L} = \frac{1}{2} \langle (u_1 + u_2)^2 \rangle dh. \quad (15)$$

The quasistatic reference in Fig. 2 integrates this generalized force with $C = C_{\text{ss}}(h)$ at each h . With unequal baths it is a steady-state path integral, not a free-energy difference. The script records the work and both mutual informations at each ramp endpoint.

References

- [1] C. Yang, X. Wei, J. Sheng, and H. Wu, "Phonon heat transport in cavity-mediated optomechanical nanoresonators," *Nature Communications* **11**, 4656 (2020). [doi:10.1038/s41467-020-18426-4](https://doi.org/10.1038/s41467-020-18426-4).
- [2] C. Yang *et al.*, Supplementary Information to *Nature Communications* **11**, 4656 (2020), Notes 1 and 2. [publisher PDF](#).